

StellarNest

Borderless Payments for Every Worker

Powered by Stellar · Built for ASEAN Migrant Workers & Freelancers

✦ Blockchain-Powered

⚡ Instant Settlement

🔒 AES-256 Encrypted



The System Is Broken at Every Level



Exorbitant Fees

Workers pay \$12-\$35 per transfer.
5-7% of each payment lost to fees.



2-3 Day Settlement

Emergency money takes days.
No same-day remittance option.



72M Unbanked in ASEAN

Recipients without a bank account
simply cannot receive transfers.



No Credit History

Informal remittances build no
financial identity or credit profile.



Fragmented Corridors

Every corridor (ID→PH, ID→VN)
is a different fee, aggregator, delay.

Introducing StellarNest

One link. Any bank. Zero crypto knowledge required.



Magic Claim Links

Shareable payment URLs via WhatsApp or SMS. Recipient claims with just a phone number — no app, no wallet, no onboarding.



Split-Routing

Every payment is automatically split: 70% to family, 30% to savings — enforced at the protocol level on the Stellar ledger.



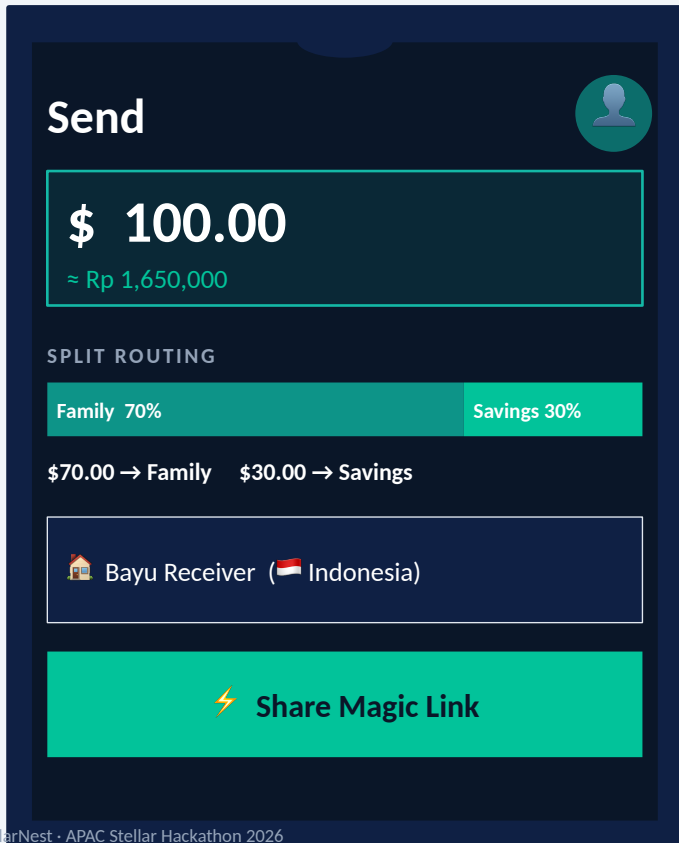
Multi-Currency Rails

BE-FAST, InstaPay, VietQR, PromptPay, DuitNow, PayNow — 6 ASEAN corridors with local settlement networks.



Client-Side AES-256-GCM Encryption · PBKDF2 (100,000 rounds) · Secrets never touch a server

The Sender Experience



StellarNest • APAC Stellar Hackathon 2026

[Screenshot: src/pages/Send.tsx]

1

Enter Amount

Type any amount (\$1–\$10,000). Live conversion shows local currency equivalent using exchange rate.

2

Set the Split

Drag the 70/30 slider. Family savings vs emergency fund — enforced on-chain in a single transaction.

3

Recipient Details

Enter name + select country. Determines settlement rail and e-wallets shown to recipient.

4

Generate & Share

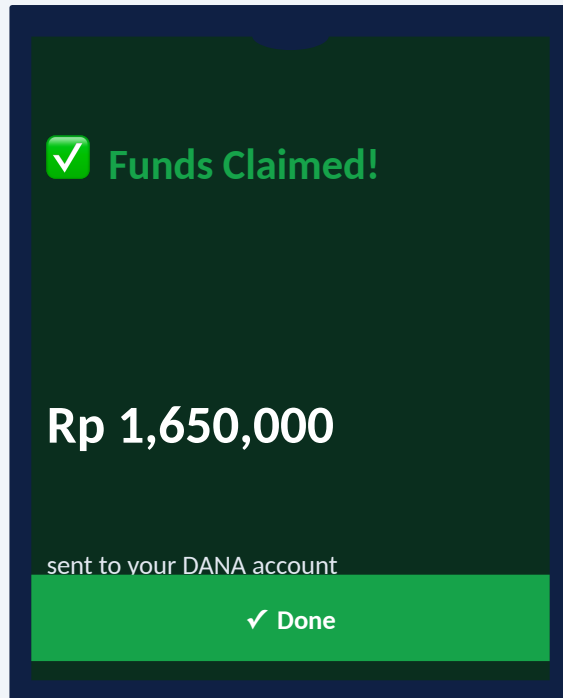
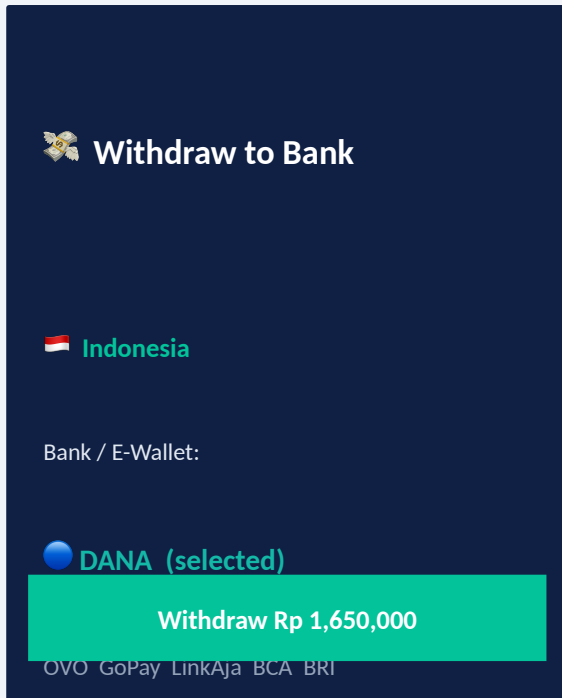
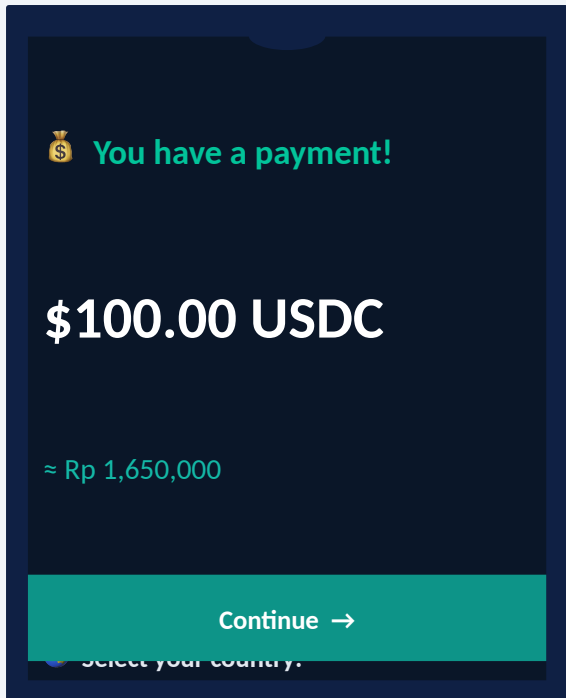
Magic Link URL created and shared via WhatsApp, SMS, or email. One tap away.

5

On-Chain

Single USDC transaction submitted to Stellar. Memo CLAIM:<id> links ledger to Firestore record.

The Recipient Experience



Step 1: Claim Page

Step 2: Withdraw

Step 3: Success!



PH  [src/pages/Claim.tsx](#)

[src/pages/Withdraw.tsx](#)



Arrives in 1-3 business days

No wallet needed · No app install · No crypto knowledge required · KYC handled by e-wallet partner

Account / Phone:

Building a Financial Identity on-Chain



[Screenshot: src/pages/Insights.tsx]



PROOF OF INCOME — StellarNest

Worker: Asep Sender | Key: GDEMOACCOUNT...

Period: January 1 — July 14, 2026

Jan 15 +\$500.00 → Family
Mar 02 +\$750.00 → Family
Apr 10 +\$600.00 → Family
May 22 +\$500.00 → Family
Jul 08 +\$300.00 → Family

TOTAL YTD: \$2,650.00



Path to Credit — verified on-chain income unlocks loans & mortgages



Auto-Generated Statement

One-click PDF from on-chain data. Verifiable by any bank.



Credit Application

Bank queries on-chain income via read-only API key (worker consents).



Micro-Loan Approval

Consistent 6-month history = auto-approved loans up to \$500.

Stack: React 19 • Firebase • Stellar SDK v16

FRONTEND



React 19 SPA

(Vite dev server)



AppContext + Services

claim.service, stellar.ts



Insights / Dashboard

YTD, charts, proof of income



DATA & SYNC



localStorage

Offline-first, sync-first writes

Firebase Firestore

Claims, profiles, async sync

AES-256-GCM Keys

Encrypted secrets, PBKDF2



STELLAR LAYER



Stellar Testnet

horizon-testnet.stellar.org

USDC Issuer

GBBD47IF6LWK7P7MDEVSCWR7...

Memo: CLAIM:<id>

Max 28 bytes (SDK v16)



Tech Stack

React 19

Frontend SPA

React Router 7

Client routing

Vite 8

Build & HMR

Firebase 12

Auth + Firestore

stellar-sdk 16

Horizon API

Tailwind CSS 3.4

Design system

Web Crypto API

AES-GCM crypto

Playwright

E2E testing



MOCK_MODE=true (dev) / false (production)

AES-256-GCM • PBKDF2 100K • Zero Server Exposure

1



Firestore UID

Anonymous Auth gives each user a unique, stable UID.

e.g. "abc123xyz"

2



PBKDF2 Key Derivation

UID + 100,000 rounds SHA-256 → 256-bit AES key.

NIST SP 800-132

3



AES-256-GCM Encrypt

Random IV per operation.
Authenticated encryption.

NIST SP 800-38D

4



Store in Firestore

base64(salt[16] || iv || M[12])
stored as encryptedSecret.

Unreadable without UID

5



In-Memory Only

Decrypted secret lives in
walletSecretRef (React useRef).
Never serialized.

Gone on sign-out



Security Specification

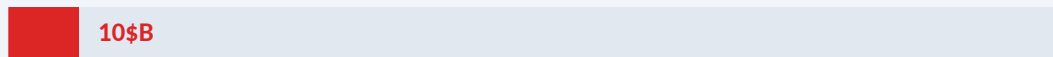
Algorithm	AES-256-GCM	NIST SP 800-38D
Key Derivation	PBKDF2-SHA256	NIST SP 800-132
PBKDF2 Rounds	100,000	OWASP 2023 recommendation
Salt / IV	128-bit / 96-bit	Unique per encryption
Secret Storage	Firestore (encrypted)	Never in localStorage
In-Memory	React useRef	Cleared on sign-out

\$150B Market. One Percent Captured = \$1.5B.

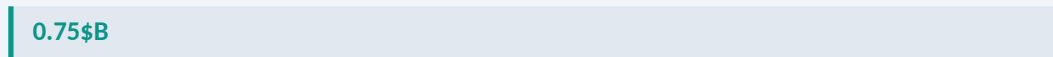
Total ASEAN Remittances



Current fees paid (5-7%)



StellarNest target (0.5%)



REVENUE STREAMS

PRIMARY



Transaction Fee (0.5%)

Applied on sender side. Recipient receives full amount. $\$500 \times 10 \text{ tx} \times 10\text{K users} = \250K/mo.

B2B



Anchor Corridor Fee

Stellar membrane partners (Flip, Sakto, VietinBank) pay per settled tx. StellarNest takes 30% revenue share.

UPSELL



Micro-Loan Origination

3% fee on approved loans. On-chain income history drives 30-second auto-approval for $\leq \$500$ loans.

From Testnet MVP to ASEAN Infrastructure

NOW	TESTNET MVP	Q3 2026	MAINNET LAUNCH	2027	MICRO-LOAN BETA	2028+	ASEAN SCALE
>	✓ USDC testnet live	>	🏛️ 5 corridors live	>	👛 2 bank partners (BCA, BRI)	>	🌐 10+ countries, 20+ corridors
>	✓ Magic Claim Links	>	🏛️ Flip, Sakti, VietinBank anchors	>	💰 Loan origination beta	>	💰 \$5M+ ARR
>	✓ 70/30 Split-Routing	>	📄 Proof of Income API	>	🌐 Thailand + Malaysia live	>	🏛️ Central bank DSP integration
>	✓ 5 ASEAN corridors	>	🔒 Security audit (Trail of Bits)	>	🇮🇩 Income API v2.0	>	🚀 1M+ active users
>	✓ Demo mode (\$2,650 YTD)	>	📱 React Native mobile app	>	🏛️ 5 more bank partners	>	🇸🇲 SEP-24/31 native support

StellarNest isn't just a payments app. It's a financial identity system built on the most efficient, most inclusive blockchain network in the world.

